

List Comprehensions

DM580: Functional Programming

Casper Bach

Reading Material

This lecture relates to these chapters of *Programming in Haskell, 2nd ed.*

- Chapter 5

Learning Objectives

- Explain what a list comprehension is
- Write simple Haskell programs involving list comprehensions

Why Comprehend List Comprehension?

The What and Why of List Comprehension

- Simple but powerful way of defining many functions on lists
- Inspired by set notation in mathematics
- Idea: specify *what* the list should contain, rather than *how* to compute it
- Derive the *how* from the *what*



Comprehending List Comprehension

Set Builder Notation

As known from mathematics

- Say I want to specify the set of all even natural numbers
- We could enumerate them; e.g., $\{2, 4, 6, 8, \dots\}$
- But that leaves it up to the reader to infer what the pattern is
- What is the pattern for $\{(1, 1), (1, 2), (2, 1), (2, 2), \dots\}$? Could mean the set of all ordered pairs of positive natural numbers, or something else.
- Set builder notation makes patterns precise

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- Precise specifications for even numbers using set builder notation:

$$\left\{ n \in \mathbb{N} \mid \text{mod } 2 = 0 \right\}, \text{ or } \left\{ 2k \mid k \in \mathbb{N} \right\} = \{0, 2, 4, \dots\}$$

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“such that”

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- The set of all square numbers

$$\left\{ n^2 \mid n \in \mathbb{N} \right\} = \{0, 1, 4, 9, \dots\}$$

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- The set of ordered pairs of nats

$$\left\{ (n, m) \mid n, m \in \mathbb{N} \right\}$$

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$[(n, m) \mid n \leftarrow [0..], m \leftarrow [0..]]$

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- For example, compare (using GHCi):

$$[(n, m) \mid n \leftarrow [0..3], m \leftarrow [0..3]] \\ = [(0,0), (0,1), (0,2), (0,3), \dots]$$
$$[(n, m) \mid m \leftarrow [0..3], n \leftarrow [0..3]] \\ = [(0,0), (1,0), (2,0), (3,0), \dots]$$

Exercise: List Comprehensions with Guards

- Build the list of all integer divisors of a number
 - E.g., for 5 it is [1,5]; for 10 it is [1,2,5,10]; for 12 it is [1,2,3,4,6,12]
- Build the list of all prime numbers
 - E.g., the list of numbers whose only divisors are 1 and the number itself



Many more examples in
the book, Chapter 5

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Questions?